# *Non-Functional Requirements*

## Product Requirements

## User Interface Requirements

The user interface of the Travel Management System should be web-based and accessible through modern browsers on both desktop and mobile devices. The interface should be responsive, consistent, and easy to navigate. Different users should have different interface views according to their roles in the system.

The user interface can be divided into the following main interfaces:

<u>Traveler Interface</u>

Which will contain:

- A header section containing the platform logo, user name, and navigation menu.
- A login and registration section where the traveler can enter credentials or create a new account.
- A travel package browsing section where the traveler can search packages by destination, price, or travel dates.
- A package details page where the traveler can view full information about a selected travel offer.
- A reservations section where the traveler can view current and past bookings.
- A feedback and review section where the traveler can leave ratings and comments.
- A follow option that allows the traveler to receive updates and promotions from travel agencies.
- A log out option that terminates the current session and redirects the user to the main page.
- Forms and actions should display validation, error, and confirmation messages when necessary.

<u>Travel Packages Interface</u>

Which will contain:

- A header section containing the platform logo, agency name, and navigation menu.
- A login section where the agency can enter credentials to access the system.
- A package management section where the agency can publish, update, or remove travel packages.
- A bookings section where the agency can view reservations made by travelers.
- A reservation management option allowing the agency to confirm bookings.
- A reviews section where the agency can view customer feedback and ratings.
- A package performance section where the agency can track the activity or visibility of their offers.
- A log out option that terminates the current session and redirects the user to the main page.

- The interface should display confirmation and error messages for actions such as adding, editing, or deleting packages.

Platform Administrator Interface

Which will contain:

- A header section containing the platform logo, administrator name, and navigation menu.
- A login section where the administrator can securely access the system.
- A user management section where the administrator can manage traveler, agency, and business accounts.
- An agency verification section where the administrator can approve or reject travel agencies.
- A monitoring dashboard where the administrator can observe system activity.
- A reports section where the administrator can generate reports related to bookings and platform usage.
- A moderation section where reported issues can be reviewed.
- A log out option that terminates the current session and redirects the user to the main page.
- The administrator interface should display structured tables, lists, and system status information for efficient monitoring.

B2B Client Interface

Which will contain:

- A header section containing the platform logo, company name, and navigation menu.
- A login and registration section where the business client can enter credentials or register using company details and tax ID.
- A verification status section where the business client can track approval of their account.
- A travel offers section where the client can browse travel packages with special B2B pricing.
- A booking section where the business client can reserve travel packages at business rates.
- A log out option that terminates the current session and redirects the user to the main page.
- Error and confirmation messages displayed during registration, verification, and booking processes.

## Usability (Learnability)

The Travel Management System should be designed to be easy to learn and use for all types of users, including travelers, travel agencies, administrators, and business clients. The

system should require minimal training and provide clear guidance for performing common tasks.

**Learnability**

- The system should be intuitive, allowing new users to understand basic functionalities without prior training.
- The interface should use clear labels, buttons, and navigation structures to help identify available actions.
- User interactions should follow consistent patterns across all pages to reduce confusion and improve familiarity.
- The system should provide helpful messages and guidance for users.
- In case of errors, the system should display clear and informative messages explaining what went wrong and how to correct it.
- Users should be able to learn how to perform basic tasks within a short period of time.
- This documentation can serve as a user guide to all the users of the website.

## Accessibility

The Travel Management System should be accessible to all users, regardless of their location, device, or basic usability needs. The system should ensure that users can access and use the platform efficiently without unnecessary barriers.

- The system should run on modern web browsers without requiring installation.
- The platform should work on desktop, tablet, and mobile devices.
- All users should access the system using their credentials(given they are provided to them).
- The system should be available at any time, provided there is internet access.
- The interface should use clear text, readable fonts, and proper contrast.
- Navigation should be simple and consistent across all pages.
- All forms should have clearly labeled input fields.
- The system should display clear error and confirmation messages.

## Dependability

The Travel Management System should operate reliably and remain available for users with minimal interruptions. The system should ensure stable performance, proper error handling, and support for maintenance and monitoring.

<u>Availability</u>

- The system should be available 24/7, except during scheduled maintenance.
- The system should aim for high availability (at least 95% uptime).

- Downtime should be minimized and communicated to users when possible.

Reliability

- The system should handle user actions (e.g., booking, publishing packages) without data loss.
- The system should maintain consistent data even in case of errors.
- The number of system failures should be minimal during normal operation.

Monitoring

- The system should log important events such as login attempts, bookings, and errors.
- Errors and failures should be recorded for troubleshooting.
- The system should provide basic monitoring of system activity and performance.

Maintenance

- The system should be designed in a flexible way to allow easy updates and fixes.
- Maintenance operations should not significantly disrupt user access.
- System updates should be manageable without affecting core functionalities.

Integrity

- The system should ensure that stored data is accurate and not corrupted.
- Unauthorized changes to data should be prevented.
- Data consistency should be maintained across all system operations.

## Errors

The system shall minimize user and system errors and provide clear ways to detect and correct them.

- The system shall display clear error messages when invalid or incomplete data is entered.
- Error messages shall explain what went wrong and how to fix it.
- The system shall validate all input fields before processing actions (e.g., login, booking, package creation).
- Users shall be able to correct errors immediately without restarting the process.

- Critical operations (e.g., booking, deleting packages) shall require confirmation to reduce mistakes.
- The system shall prevent duplicate or invalid submissions where possible.

## Satisfaction

The system shall provide a positive and efficient user experience for all users.

- The interface shall be simple, clean, and easy to use.
- Users shall be able to complete common tasks quickly.
- The system shall provide clear feedback after every action (success or failure).
- Navigation shall be intuitive and consistent across all pages.
- The system shall reduce unnecessary steps when performing actions.

## Security

The Travel Management System shall protect user data and prevent unauthorized access, modification, or misuse.

- The system shall require authentication (username/email and password) for all users.
- The system shall enforce authorization, ensuring users can only access features related to their role.
- Passwords shall be stored in encrypted form.
- The system shall use secure communication protocols (e.g., HTTPS).
- Sensitive data shall be protected from unauthorized access.
- The system shall log important actions such as login attempts and critical operations.
- The system shall prevent unauthorized access through basic security mechanisms (e.g., input validation, session management).